

Pandas Date & Time Cheat Sheet

1. Convert to datetime

```
pd.to_datetime(df["date"])
pd.to_datetime(df["date"], format="%Y-%m-%d")
pd.to_datetime(df["date"], errors="coerce")
```

2. Common format strings

```
%Y = 2025
%y = 25
%m = 01-12
%d = 01-31
%H = hour (24h)
%M = minute
%S = second
```

Examples:

```
2025-07-16 -> %Y-%m-%d
16/07/2025 -> %d/%m/%Y
07/16/2025 -> %m/%d/%Y
20250716 -> %Y%m%d
2025-07-16 14:30:15 -> %Y-%m-%d %H:%M:%S
```

Extract Date Components

```
df["year"] = df["date"].dt.year
df["month"] = df["date"].dt.month
df["day"] = df["date"].dt.day
df["quarter"] = df["date"].dt.quarter
df["weekday"] = df["date"].dt.weekday
df["day_name"] = df["date"].dt.day_name()
df["month_name"] = df["date"].dt.month_name()
df["date_only"] = df["date"].dt.date
```

Formatting

```
df["date"].dt.strftime("%Y-%m-%d")
df["date"].dt.strftime("%d/%m/%Y")
```

```
df["date"].dt.strftime("%m/%d/%Y")
```

```
df["date"].dt.strftime("%Y%m%d")
```

Filtering

```
df[df["date"] >= "2025-01-01"]
```

```
df[(df["date"] >= "2025-01-01") & (df["date"] < "2026-01-01")]
```

Sorting

```
df.sort_values("date")
```

```
df.sort_values("date", ascending=False)
```

Grouping

```
df["year"] = df["date"].dt.year
```

```
df.groupby("year")["close"].max()
```

```
df.loc[df.groupby("year")["volume"].idxmax()]
```

```
max_close = df.groupby("year")["close"].transform("max")
```

```
df[df["close"] == max_close]
```

Time Series

```
df = df.sort_values("date")
```

```
df["prev_close"] = df["close"].shift(1)
```

```
df["price_change"] = df["close"].diff()
```

```
df["daily_return"] = df["close"].pct_change()
```

```
df["rolling_mean"] = df["close"].rolling(20).mean()
```

Missing Dates

```
df.isna().sum()
```

```
df.dropna()
```

```
df.fillna(0)
```

```
df.ffill()
```

`df.bfill()`

Interview Tips

Know the difference between:

- `pd.to_datetime` vs `strptime`
- `idxmax` vs `transform("max")`
- `merge` vs `concat`
- `loc` vs `iloc`
- `agg` vs `transform`

For pair programming:

1. Inspect schema (head, columns, dtypes)
2. Convert date
3. Clarify requirements
4. Implement
5. Discuss edge cases